

Deciphering the roles of leucine-rich repeat receptor-like protein kinases (LRR-RLKs) in response to *Fusarium* wilt in the *Vernicia fordii* (Tung tree)

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ABSTRACT

Leucine-rich repeat receptor-like protein kinases (LRR-RLKs) are vital for plant growth and development, signal transduction, immunity, and play diverse roles in plant defense responses. However, the LRR-RLK genes have not been systematically studied in *Vernicia fordii* (tung tree), especially its response to *Fusarium* wilt. Here, we carried out an integrative analysis of LRR-RLKs among five Euphorbiaceae species: *Hevea brasiliensis* (rubber tree), *Manihot esculenta* (cassava), *Jatropha curcas* (physic nut), *Ricinus communis* (castor bean), and *V. fordii*, which contained 223, 311, 186, 138, and 167 LRR-RLKs, respectively. Maximum-likelihood tree was estimated using LRR-RLKs of *Arabidopsis thaliana* as a template, and they allowed us to divide Euphorbiaceae LRR-RLKs into 22 groups. There are 126 segmental and 30 tandem duplications in these Euphorbiaceae genomes by synteny analysis. The tissue-specific expression patterns revealed that *V. fordii* LRR-RLKs (VfLRR-RLKs) were differentially expressed in various tissues, and some of them exhibited specific expression in meristems tissues, which suggested their potential functions during organ formation and cell fate specification. Two VfLRR-RLK pairs (Vf01G2125 and Vf03G1740, Vf06G2687 and Vf10G1659), which generated by tandem duplication events, were associated with possible resistance to *Fusarium* wilt infection. The qRT-PCR confirmed these four VfLRR-RLKs contained opposite expression profiles during pathogen infection in *V. fordii* and *V. montana*. Taken together, our data systematically analyzed the LRR-RLK family in Euphorbiaceae genomes for the first time. We highlight the putative roles of VfLRR-RLKs in response to *Fusarium* wilt infection, and VfLRR-RLKs may be further applied in marker-assisted breeding to control *Fusarium* wilt in *V. fordii*.

1. Introduction

During growth and development, a series of different signals of cell-environment and cell-cell interaction can stimulate plants to induce a variety of different kinds of signal transduction. The regulatory mechanism for reversible phosphorylation associated with protein kinases (PKs) contributes to the process of cell signal transduction. In plants, all living cells can process and sense signals of internal or external through

cell surface PKs, most of which are from receptor-like kinase (RLK) family members (Gou et al., 2010; Shiu and Bleecker, 2001a). In 1990, Walker and Zhang (1990) identified the first plant leucine-rich repeat receptor-like kinase (LRR-RLK) in maize (*Zea mays*) and since then, many LRR-RLKs have been isolated from other plant genomes (Fischer et al., 2016; Walker and Zhang, 1990; Yuan et al., 2018). The structural characteristics of LRR-RLK proteins contain three domains: a conserved cytoplasmic threonine/serine kinase (KD), an extracellular variable

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receptor, and a single-pass transmembrane (Walker, 1994).

The *LRR-RLKs* can be divided into two categories in plants (Diévert and Clark, 2003). Firstly, the *LRR-RLKs* are vital for hormone signaling, organogenesis, and morphogenesis, indicating that they may regulate plant growth and development. Secondly, multiple members of *LRR-RLK* can respond to abiotic and biotic stresses such as *Fusarium* wilt, cold, salt, and drought, and therefore may be related to defense (Afzal et al., 2008; Cao et al., 2020a; Godiard et al., 2003). Among them, some *LRR-RLKs* have been confirmed to contain dual functions, which may be due to the crosstalk between developmental pathways and defense or the recognition of multiple ligands by one signal receptor (Afzal et al., 2008). For example, some studies have confirmed that *ERECTA* contributes to resist bacterial wilt and ovule development (Diévert and Clark, 2004; Godiard et al., 2003; Torii et al., 1996). Although the research on the function of *LRR-RLKs* has been made great progress in recent years, there are still open questions for most *LRR-RLKs* (Diévert and Clark, 2004). Diévert and Clark (2004) found that *LRR-RLKs* are vital for various growth and development processes by observing the phenotype of its *Arabidopsis thaliana* mutants. At the same time, some members of the *LRR-RLK* family are made more complicated by the functional redundancy, sub-functionalization, or neo-functionalization of their signaling network. For instance, in shoot apical meristem, the expression profile of the stem cell-promoting *WUSCHEL* (*WUS*) transcription factor regulated by a receptor complex formed with *CLAVATA1* (*CLV1*) and *CLV2* upon the perception of the *CLV3*/*ESR*-related (*CLE*) peptide derived from *CLV3* (Brand et al., 2000; Mayer et al., 1998; Schoof et al., 2000). Additional receptors, such as Barely any Meristem (*BAM1*, -2, and -3), have high sequence similarity with *CLV1* but implement seemingly contradictory functions (DeYoung and Clark, 2008). *CLV1* plays a crucial role in promoting the differentiation of stem cells, and *BAM* receptors are essential for maintaining stem cells (DeYoung and Clark, 2008). The differences in genetic function between *CLV1* and *BAM* receptors are largely driven by their distinctive expression patterns, although they maintain significant similarities in biochemical functions, as described by DeYoung and Clark (2008).

Besides *A. thaliana*, *LRR-RLKs* have been studied and investigated in several species, such as *Solanum tuberosum*, *Solanum lycopersicum*, *Vitis amurensis*, *Pyrus bretschneideri*, and *Oryza sativa* (Fischer et al., 2016; Li et al., 2018; Magalhães et al., 2016; Sakamoto et al., 2012; Shiu et al., 2004; Sun et al., 2017; Zan et al., 2013), while their functions could be divergent or conserved. *Vernicia fordii* (Hemsl.) Airy Shaw (Euphorbiaceae), commonly known as Tung tree, is of great economic significance (Cui et al., 2018; Zhang et al., 2019), but its growth and development are affected by a large number of phytopathogens (Zhang et al., 2018). Among them, *Fusarium oxysporum* f. Sp. *Fordii* (*Fof-1*) disease has the most serious effect on the growth and development in *V. fordii*. On the contrary, *V. montana* showed significant resistance to *Fusarium* wilt disease (Zhang et al., 2018). Considering the recent genome sequencing of *Hevea brasiliensis* (rubber tree), *Manihot esculenta* (cassava), *Jatropha curcas* (physic nut), *Ricinus communis* (castor bean), and *V. fordii* (Bredeson et al., 2016; Chan et al., 2010; Cui et al., 2018; Ha et al., 2019; Liu et al., 2020; Wu et al., 2015; Zhang et al., 2019), and the involvement of *LRR-RLK* genes in defense responses, we carried out a systematic analysis to clarify the evolutionary relationships, duplication events, and expression patterns. Especially, we compared the transcriptional activity after *Fusarium* wilt infection between *V. fordii* and *V. montana*, and finally identified four *V. fordii* *LRR-RLKs* (*VfLRR-RLKs*) genes that might be involved in the resistance of *V. fordii* to *Fusarium* wilt.

2. Results and discussion

2.1. Identified of *LRR-RLKs*

The genome data of *M. esculenta*, *H. brasiliensis*, *J. curcas*, *R. communis*, and *V. fordii* were recently released, respectively (Bredeson et al., 2016; Chan et al., 2010; Cui et al., 2018; Ha et al., 2019; Liu et al.,

2020; Wu et al., 2015; Zhang et al., 2019). To determine the *LRR-RLK* gene family members, a combined computational approach was used in these tested Euphorbiaceae genomes (Cao et al., 2019b, 2020a). Finally, 1025 *LRR-RLKs* were identified, including 311 in *M. esculenta*, 223 in *H. brasiliensis*, 186 in *R. communis*, 138 in *J. curcas*, and 167 in *V. fordii*, respectively (Table 1 and Table S1). Fischer et al. (2016) also identified *LRR-RLKs* in *M. esculenta*, *J. curcas*, and *R. communis* (Fischer et al., 2016), and the members of *LRR-RLK* family were different from those in our research, possibly due to the different genome versions used. In the previous study, the genomes of *M. esculenta*, *J. curcas*, and *R. communis* were not assembled into chromosomes, while our study used the latest version of high-quality genomes. At the same time, the Pfam (Punta et al., 2011), InterProScan (Jones et al., 2014), and CDD (Marchler-Bauer et al., 2012) were also used to identify the functional domain of each protein, so the member of *LRR-RLK* family members identified in this study should be more accurate.

Compared with other gene families, the *LRR-RLK* gene family is a relatively large family in plants. For example, 427 *LRR-RLK* genes have been identified in *Pyrus bretschneideri*, 379 in *Populus trichocarpa*, 300 in *Citrus clementina*, 236 in *A. thaliana*, and 226 in *Zea mays* (Li et al., 2018; Magalhães et al., 2016; Sakamoto et al., 2012; Shiu et al., 2004; Sun et al., 2017; Zan et al., 2013). Furthermore, the sizes of the genome not directly related to the *LRR-RLK* family members. For example, the number of *LRR-RLKs* in *M. esculenta* (337 members) and *H. brasiliensis* (342 members) has not changed obviously, but their genome size has changed significantly. On the contrary, the genome size of the *R. communis* (305 Mb) and *J. curcas* (329 Mb) has a corresponding relationship with the number of their *LRR-RLKs*. The number of *LRR-RLKs* in *M. esculenta* and *H. brasiliensis* almost doubled compared with that in *R. communis*, *J. curcas*, and *V. fordii*. The chromosome

Table 1
Total number of receptors distributed in the different groups of *LRR-RLKs*.

Groups	Plant species	<i>J. curcas</i>	<i>R. communis</i>	<i>M. esculenta</i>	<i>H. brasiliensis</i>
	<i>V. fordii</i>				
LRR I	7	1	7	11	12
LRR II	10	9	10	16	22
LRR III	32	19	32	45	41
LRR IV-1	3	8	11	8	17
LRR IV-2	3	2	1	5	6
LRR V	6	6	6	8	9
LRR VI-1	4	3	6	7	8
LRR VI-2	3	1	4	7	9
LRR VII-1	2	1	2	3	2
LRR VII-2	5	5	6	8	7
LRR VIII	2	2	3	7	5
LRR IX	7	4	5	7	8
LRR Xa	9	2	3	9	8
LRR Xb-1	0	0	1	1	2
LRR Xb-2	4	7	7	10	5
LRR XI-1	1	0	1	2	0
LRR XI-2	19	38	37	74	45
LRR XII	41	21	34	70	4
LRR XIIIa	2	1	2	3	3
LRR XIIIb	2	2	2	3	4
LRR XIV	2	3	2	3	4
LRR XV	3	3	4	4	4

numbers of *R. communis*, *J. curcas*, and *V. fordii* are 22, 20, and 22, respectively, whereas those of the *M. esculenta* and *H. brasiliensis* are 38, suggesting that the members of *LRR-RLK* gene family have experienced expansion corresponding to the variation in the number of chromosomes. Remarkably, both whole-genome duplication (WGD) events are shared by *M. esculenta* and *H. brasiliensis* but *R. communis*, *J. curcas*, and *V. fordii* do not share the recent WGD event (Bredeson et al., 2016; Chan et al., 2010; Cui et al., 2018; Ha et al., 2019; Liu et al., 2020; Wu et al., 2015; Zhang et al., 2019), which probably contributed to the expansion of *LRR-RLK* members in *M. esculenta* and *H. brasiliensis*.

2.2. Evolutionary analyses of *LRR-RLKs*

To determine the evolutionary history of *LRR-RLKs* in Euphorbiaceae genomes, the currently accepted genome assemblies were used to find, classify, and carry out a systematic genomic study for the *LRR-RLKs* in this work. The ML tree showed that all *LRR-RLKs* in Euphorbiaceae could be divided into different groups (Fig. 1 and Table 1), which were determined and defined based on the *A. thaliana* homologs

nomenclature within the same group (Shiu and Bleeker, 2001b). Most groups in each phylogeny were well-supported (Fig. 1), suggesting the reliability of the ML tree. Interestingly, some *LRR-RLKs* revealed species-specific characteristics due to the high degree of duplication in correspondent genomes. For example, eight *A. thaliana* *LRR-RLKs* (AT2G24130.1, AT3G47090.1, AT3G47110.1, AT3G47570.1, AT3G47580.1, AT5G20480.1, AT5G39390.1, and AT5G46330.1) in the group XII, while 4 *H. brasiliensis* *LRR-RLKs* (HbLRR-RLKs), 22 *J. curcas* *LRR-RLKs* (JcLRR-RLKs), 34 *R. communis* *LRR-RLKs* (RcLRR-RLKs), 41 *VfLRR-RLKs*, and 70 *M. esculenta* *LRR-RLKs* (MeLRR-RLKs) were found as the orthologous genes of these eight *A. thaliana* *LRR-RLKs*. The rapid loss events of *LRR-RLKs* were identified in group XII between *J. curcas* and *H. brasiliensis*, as well as the rapid expansion of *LRR-RLKs* events were found in group XII among *R. communis*, *V. fordii* and *M. esculenta* (Table 1). These results suggested that the gene duplication and loss events might play significant roles in the expansion and contraction of *LRR-RLK* family members (Fig. 1 and Table S2).

As previously reported, most *A. thaliana* *LRR-RLKs* similar functions tend to cluster together (Shiu and Bleeker, 2001b), so the homologous

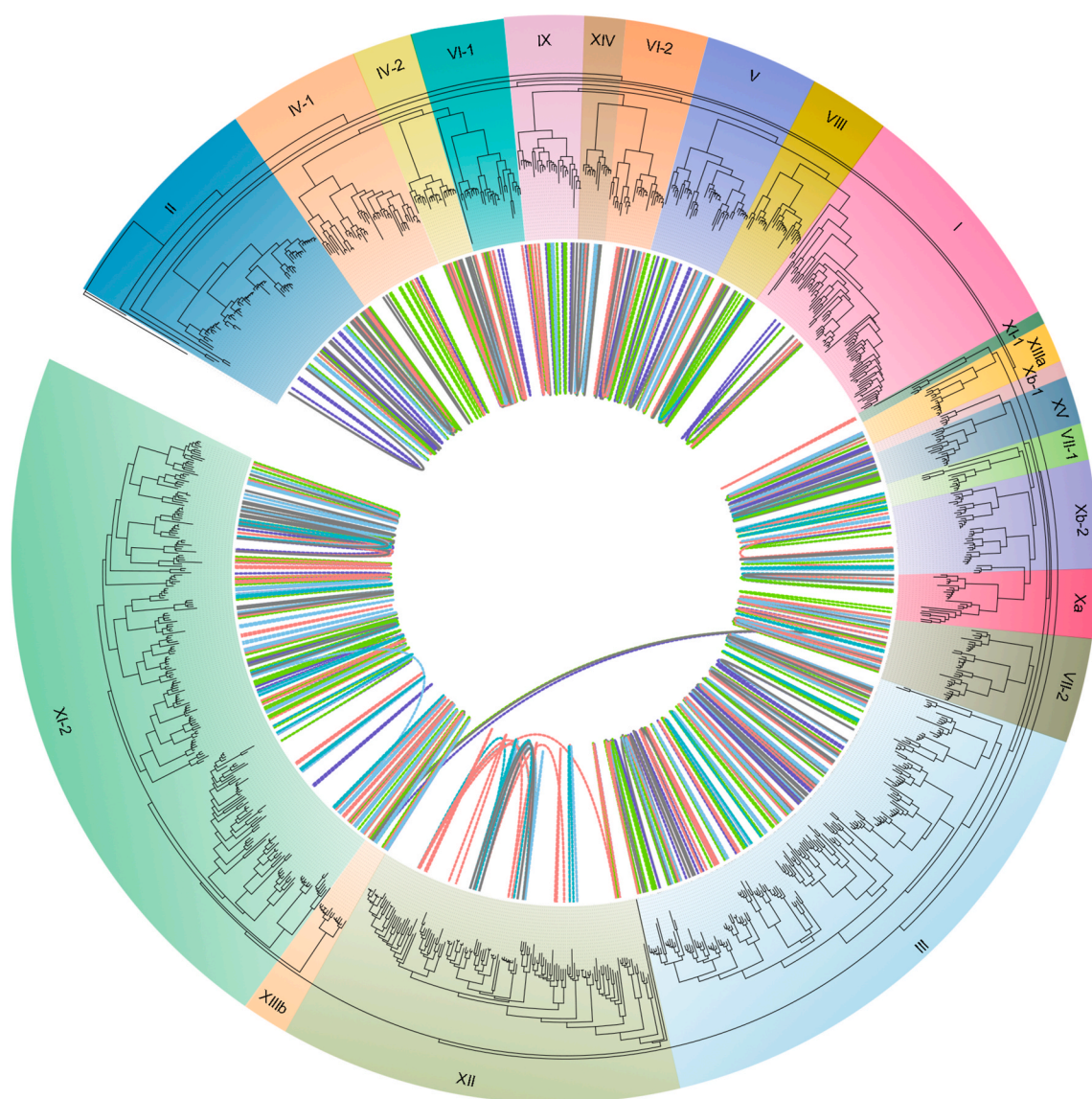


Fig. 1. The maximum likelihood tree and synteny analysis among these five Euphorbiaceae genomes. All *LRR-RLKs* were divided into 22 groups and were distinguished by different colors. These different groups were determined and defined based on the *A. thaliana* homologs nomenclature within the same group (Shiu and Bleeker, 2001b). The synteny relationships between different Euphorbiaceae genomes were represented by different links. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

genes of Euphorbiaceae in the same group with *A. thaliana* LRR-RLKs might have similar functions. In *A. thaliana*, at least one LRR-RLK from each group has been functionally characterized, except for groups VIII and IV that do not contain *A. thaliana* orthologous genes with identified functions. For example, the *A. thaliana* LRR-RLKs from group V were associated with root hair specification (Dolan, 2006; Eý¹oglu et al., 2007; Tanaka et al., 2009). Group X contained the *A. thaliana* LRR-RLKs that played key roles in peptide signaling fell and brassinosteroid signal transduction (Kinoshita et al., 2005). Group XI included *A. thaliana* LRR-RLKs associated with defense response, abscisic acid signaling, vascular development, organ morphogenesis, and cell fate specification (Sánchez-Rodríguez et al., 2009; van Zanten et al., 2009; Wang et al., 2011). Additionally, *ERECTA-LIKE* and *ERECTA* genes from group XIII-b can regulate stomatal development (Shpak et al., 2003; Torii et al., 1996; Wu et al., 2016), while two *FEI* genes from group XIII-a are related to the cell wall development signaling pathway (Xu et al., 2008). The Euphorbiaceae LRR-RLKs in group I, -II, -III, -VII, and -XII were clustered with *A. thaliana* LRR-RLKs might play important roles in inducing defense signals and tissue development.

2.3. Contributions of segmental and tandem duplications to the LRR-RLKs family size

Four processes of genetic evolution, including transposition, tandem duplication (TD), chromosome fragment insertion mutation, and chromosome doubling, can generate novel gene functions (Cao et al., 2019a, 2019b; Samonte and Eichler, 2002). Previous studies shown that Euphorbiaceae species have experienced at least one WGD; the recent WGD was shared only by both *M. esculenta* and *H. brasiliensis*, which occurred about 30–45 million years ago (MYA) corresponding to

synonymous substitution rate (Ks) between 0.23 and 0.37; the ancient WGD was shared by all five examined species, which occurred about 140 MYA corresponding to Ks between 1.08 and 1.48 (Bredeson et al., 2016; Chan et al., 2010; Cui et al., 2018; Ha et al., 2019; Liu et al., 2020; Wu et al., 2015; Zhang et al., 2019). Subsequently, we executed an intra-genome scan of collinearity blocks to trace which LRR-RLKs evolved from these two WGDs (Fig. 2 and Table S3). The calculated Ks value of each pair revealed that these pairs traced to the ancient WGD were identified in *J. curcas* (10 JcLRR-RLKs), *R. communis* (6 RcLRR-RLKs), and *V. fordii* (17 VfLRR-RLKs), while gene pairs traced this WGD were not identified in *H. brasiliensis* and *M. esculenta*. Subsequently, gene pairs traced to the recent WGD were only identified in *M. esculenta* and *H. brasiliensis*. In total, 8 gene pairs in *M. esculenta* and 24 pairs in *H. brasiliensis* were determined to be related to the recent WGD, implying that this WGD played an important role in the expansion of the LRR-RLKs in these genomes. Among each species related to segmental duplication/WGD, the proportion of LRR-RLKs ranged from 11% in *M. esculenta* to 27% in *R. communis*. All groups were distributed with duplicated gene pairs, but some of them were lineage-specific (Figs. 1 and 2, and Table S3). For example, XIIIa and XIIIb only found WGD/segmental duplication in *H. brasiliensis*, and XIV only found it in Euphorbiaceae, but not in *A. thaliana*; these results provide new insights and clues to the evolution of different groups in Euphorbiaceae.

Tandem duplication (TD) can increase the genome complexity and gene numbers, which is another way to expand the members of the gene family (Cao et al., 2016; Lu et al., 2012). Therefore, we compared the number of TD genes in each Euphorbiaceae genome to explore the important role of TD in the expansion of the LRR-RLK gene family. There are 8 in *M. esculenta*, 5 in *J. curcas*, 10 in *R. communis*, 2 in *V. fordii*, and 5 TD gene pairs *H. brasiliensis* (Table S4). These involved 4 VfLRR-RLKs in

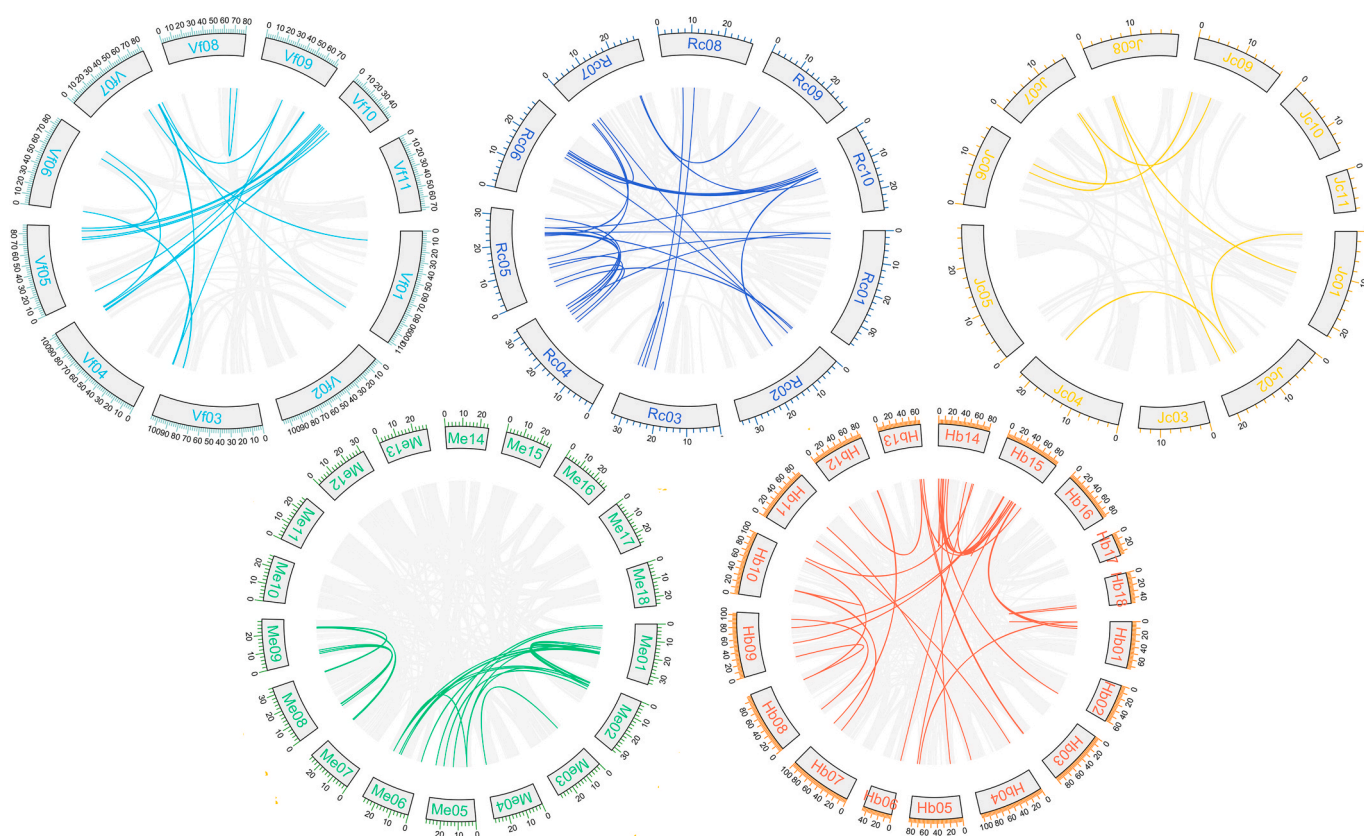


Fig. 2. The circo figures for chromosome locations with segmental duplication links in *M. esculenta* (Me; green), *H. brasiliensis* (Hb; orange), *J. curcas* (Jc; yellow), *R. communis* (Rc; blue), and *V. fordii* (Vf; cyan). The different lines suggested segmented duplicated gene pairs among these five Euphorbiaceae genomes. All the collinearity pairs are represented by grey background. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

V. fordii, 16 RcLRR-RLKs in *R. communis*, 8 JcLRR-RLKs in *J. curcas*, 15 MeLRR-RLKs in *M. esculenta*, and 8 HbLRR-RLKs in *H. brasiliensis*. Previously published articles revealed that most TD events (28/37 LRR-RLKs) in *A. thaliana* have occurred in group I (Gou et al., 2010; Liu et al., 2017). Similar results also occurred in Euphorbiaceae species, indicating that TD events have different effects on specific groups. Remarkably, although the number of groups expanded by TD was different in Euphorbiaceae, the observation that some groups in all tested species (I-1, XI-1, and XII-1) were consistently expanded through TD, while others did not show a preference for TD.

2.4. Highly conserved collinearity relationships were identified between *V. fordii* and other four euphorbiaceae genomes

The distributions of all LRR-RLKs on the chromosomes were obtained in five Euphorbiaceae genomes, based on the GFF annotation files. In *V. fordii*, chr 3 (27 VfLRR-RLK genes) and chr 9 (17 VfLRR-RLK genes) contained more LRR-RLKs than the others. Ten VfLRR-RLK genes were not scattered on any chromosomes. The orthologous LRR-RLKs were detected to understand the potential evolutionary mechanisms of the LRR-RLK family in Euphorbiaceae genomes. Here, we observed the conserved collinearity relationships in these five Euphorbiaceae genomes by MCScanX software. Subsequently, 2012 syntenic blocks containing the LRR-RLK gene were found in chromosomes of these five Euphorbiaceae genomes (Fig. 3 and Table S5). Among these syntenic blocks, a total of 624 LRR-RLK genes (114 from *V. fordii*, 146 from *R. communis*, 114 from *M. esculenta*, 165 from *H. brasiliensis*, and 85 from *J. curcas*) were present. These data highlight and confirm the correspondence between syntenic blocks gene pairs and phylogenetic relationships (Fig. 1). In addition, 401 LRR-RLK genes (i.e. 53 VfLRR-RLKs, 40 RcLRR-RLKs, 197 MeLRR-RLKs, 58 HbLRR-RLKs, and 53 JcLRR-RLKs) were not present in any syntenic blocks, indicating these genes may have evolved from independent duplication events.

We determined 130 orthologous gene pairs between *V. fordii* and *H. brasiliensis*, 98 pairs between *V. fordii* and *M. esculenta*, 133 pairs between *V. fordii* and *R. communis*, 80 pairs between *V. fordii* and *J. curcas* (Fig. 3 and Table S5), indicating the LRR-RLK genes presented different evolutionary mechanisms among different species after being evolved from the same ancestor. Interestingly, we found some gene pairs between *V. fordii* and *J. curcas*, but not found between *V. fordii* and *R. communis*/*M. esculenta*/*H. brasiliensis*, implying these orthologous pairs formed after *R. communis*/*M. esculenta*/*H. brasiliensis* and evolved from a common ancestor of *J. curcas* and *V. fordii*, or these orthologous pairs are lost in *V. fordii* and *R. communis*/*M. esculenta*/*H. brasiliensis* after divergence of these species, such as Vf11G1528/Jatcu.03g000454. Additionally, we also observed many several-for-one collinearity relationships between *V. fordii* and *R. communis*/*M. esculenta*/*H. brasiliensis*/*J. curcas*, such as Vf05G1797/GT003585/GT003775 and Vf10G0524/Jatcu.09g000179/Jatcu.06g001577, further explaining the divergent expansion patterns of LRR-RLKs in different species during evolution.

2.5. Expression profiles of VfLRR-RLKs during vegetative and reproductive development

Although some manuscripts have suggested that LRR-RLKs are involved in plant growth and development (Colette et al., 2011; Shpak et al., 2003; Torii, 2004; Wu et al., 2016), to the best of our knowledge, the study of VfLRR-RLKs in *V. fordii* is still limited. To get more clues on the functions and roles of VfLRR-RLKs at different stages of reproductive development in *V. fordii*, we utilized the RNA-seq datasets generated in-house (Cui et al., 2018; Zhang et al., 2019). Overall, we found that VfLRR-RLKs exhibited distinct tissue-specific expression patterns (Fig. S1). Among them, some VfLRR-RLKs demonstrated specific transcript accumulations in roots, leaves, developing female flowers, developing male flowers, and developing seeds, such as Vf05G1941 in leaves and Vf05G0232 in roots. The identification of these VfLRR-RLKs specifically expressed in meristems tissues may provide key clues for the functions during organ formation and cell fate specification. Therefore, our data highlighted the function of VfLRR-RLKs in meristem tissues and might lay a foundation for further study on the function of these genes. Remarkably, the VfLRR-RLKs (Vf07G0170 and Vf10G1082 from group XIIIa, and Vf10G0507 and Vf08G1383 from group XIIIb) were relatively higher expressed during flower development (Fig. S1). Specifically, two expressed genes Vf10G0507 and Vf08G1383 (group XIIIb) were orthologous genes of *A. thaliana* *ERECTA* and *ERECTA-LIKE* genes, which regulate organ size and stomatal development (Godiard et al., 2003; Sánchez-Rodríguez et al., 2009; Torii et al., 1996; van Zanten et al., 2009).

In *V. fordii*, we identified two tandem duplicated gene pairs. Among them, Vf09G0613-Vf09G0614 exhibited divergent expression patterns at different stages of reproductive development in *V. fordii*, suggesting that these two genes are functionally divergent after undergoing TD (Fig. S2). Previously published manuscripts suggested that in *O. sativa*, *A. thaliana*, and *P. trichocarpa*, more than 50% of duplicated LRR-RLK pairs that were evolved from a segmental duplication/WGD showed expressional divergence. In *V. fordii*, we identified 19 segmental duplication/WGD pairs. Among them, six duplicated gene pairs shared similar expression patterns, and the remaining gene pairs exhibited expressional divergence (Fig. 4). We also noted that the amino acid sequence similarity of gene pairs sharing similar expression patterns is greater than that of other gene pairs, suggesting that the sequence variation was correlated with the expressional diversity of duplicated pairs in *V. fordii*. These data exhibited that the LRR-RLK family has dynamic diversity in the evolutionary process, which might contribute to the adaptability of trees.

2.6. Expression of VfLRR-RLKs in response to fusarium infection

As the main oil-producing species, *V. fordii* is susceptible to *Fusarium* wilt disease and seriously affects its growth (Shockey et al., 2016; Yang et al., 2014). However, *V. montana* is resistant to *Fusarium* wilt disease (Chen et al., 2016; Shockey et al., 2016; Yang et al., 2014). Members of LRR-RLK family have been confirmed to be widely involved in plant stress resistance (Chen et al., 2016; Colette et al., 2011; Tang et al.,

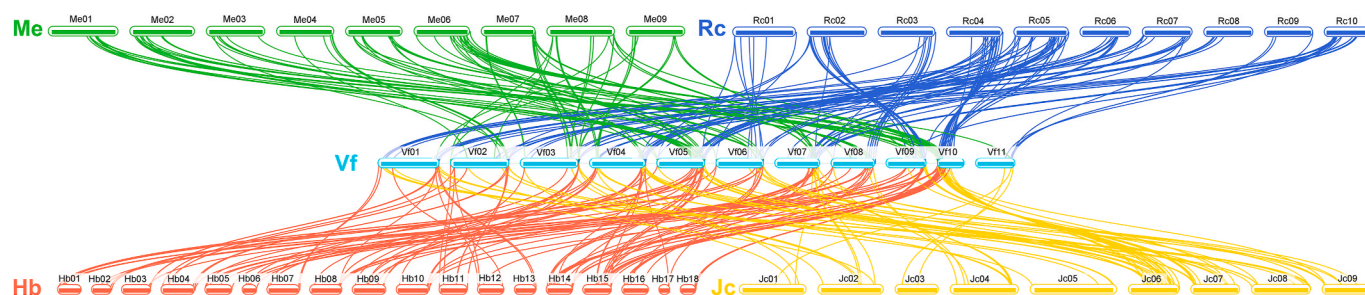


Fig. 3. Collinearity relationships of LRR-RLKs in *V. fordii* and the other four Euphorbiaceae genomes. The chromosomes of different Euphorbiaceae species were depicted as blocks of different colors. Gene pairs with a syntenic relationship between different Euphorbiaceae species were connected by different colored lines. (For interpretation of the references to color in this figure legend, the reader is referred to the Web version of this article.)

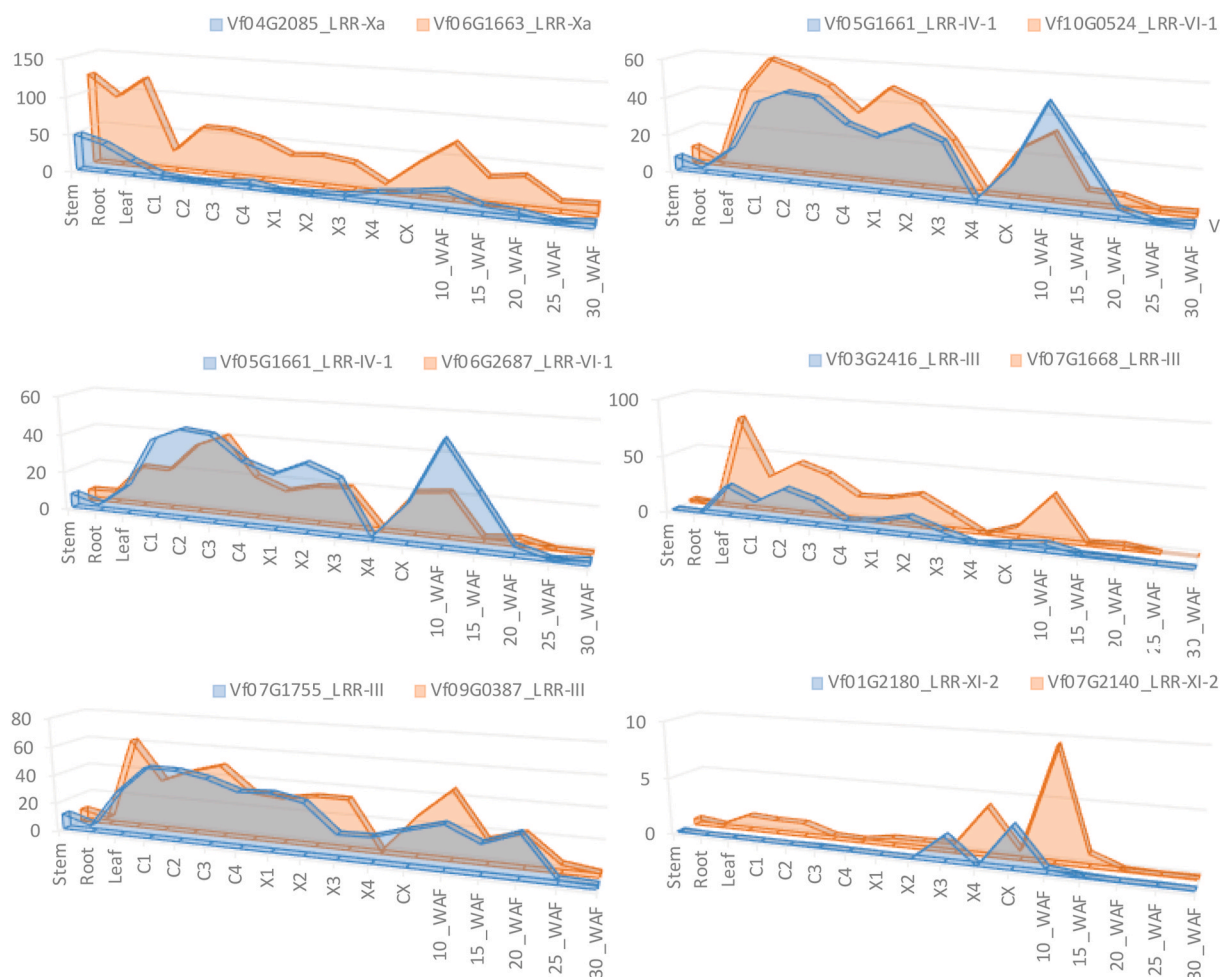


Fig. 4. The similar expression patterns between duplicated *VfLRR-RLK* gene pairs during vegetative and reproductive development. Numbers on the x-axis indicate the following: seeds at 10, 15, 20, 25, and 30 weeks after flowering (WAF; 10_WAF, 15_WAF, 20_WAF, 25_WAF, and 30_WAF), male flowers at 30, 20, 10, and 1 days before flowering (X1–X4), female flowers at 30, 20, 10, and 1 days before flowering (C1–C4), young leaves, roots, stems, and hermaphrodite (CX).

2010; Ye et al., 2017), so *LRR-RLKs* may be used as candidate genes for future genetic breeding. According to the transcriptomic analysis, we analyzed the expression patterns of *VfLRR-RLKs* in both the *Fusarium* wilt susceptible and resistant varieties (Fig. 5), including *V. montana* and *V. fordii*. The expression profiles of *LRR-RLKs* showed to vary between *V. montana* and *V. fordii* in response to *Fusarium* wilt infection. In *V. fordii* variety, 72 *VfLRR-RLKs* were significantly down-regulated at F1, F2, and F3 infection stages, but the remaining *VfLRR-RLKs* were up-regulated in one or more of these three infection stages. In the resistant *V. montana* variety, we only found that 21 *VfLRR-RLKs* were down-regulated at M1, M2, and M3 infection stages, while the remaining *VfLRR-RLKs* were up-regulated in one or more of these three infection stages (Fig. 5). These data indicated that most *VfLRR-RLKs* exhibited different expression patterns in response to *Fusarium* wilt disease in these two species (*V. montana* and *V. fordii*) (Fig. 5). Among these *VfLRR-RLKs*, four genes (i.e. *Vf01G2125* and *Vf10G1659* from group III, *Vf03G1740* and *Vf06G2687* from group VI-1) displayed sustained decreased expression at F1, F2, and F3 infection stages in *V. fordii*, while these four genes showed sustained increased expression at M1, M2, and M3 infection stages in *V. montana* (Fig. 5). The qRT-PCR experiment was used to confirm the gene expression patterns of these four *VfLRR-RLKs*. The gene-specific primers (Table S6) of these *VfLRR-RLKs* were designed using Primer-BLAST. The results of qRT-PCR analyses exhibited similar expression patterns as transcriptome (Fig. 6). These four *LRR-RLKs* (i.e. *Vf01G2125*, *Vf03G1740*, *Vf06G2687*, and *Vf10G1659*) might be the key gene that played important roles in *Fusarium* wilt resistance. Indeed,

Wang et al. (2019) found that two genes (*VmMYB041* and *VmMYB011*) help *V. montana* to resist *Fusarium* wilt, because *VmMYB041* and *VmMYB011* showed the trend of rising in M1 and then decreasing, while their corresponding orthologous genes *VfMYB029* and *VfMYB011* had a trend of decreasing first and then increasing during the infection period (Wang et al., 2019). In *A. thaliana*, Dai et al. (2019) discovered an *LRR-RLK* gene, *At5G05160* from group III, which may contribute to restrict the fungal susceptibility of the rosette core (Dai et al., 2019).

Additionally, these two gene pairs (i.e. *Vf01G2125*, *Vf03G1740*, *Vf06G2687*, and *Vf10G1659*) were generated by TD events. Further, one gene pair (*Vf01G2125* and *Vf10G1659*) had similar expression patterns. However, the other gene pair (*Vf06G2687* and *Vf10G1659*) contained asymmetric expression patterns (Fig. 7). To confirm the gene expression patterns of these *VfLRR-RLKs*, we carried out qRT-PCR experiments. The results of qRT-PCR analyses exhibited similar expression profiles as transcriptome (Fig. S3). The current data suggested that the *VfLRR-RLKs* might have different regulatory mechanisms when responding to *Fusarium* wilt infection.

3. Conclusions

We identified *LRR-RLKs* in five Euphorbiaceae genomes, and further estimated their diversity in terms of members, evolutionary relationships, duplication events, tissue expression profiles, as well as expression profiles in response to *Fusarium* wilt infection. The lineage-specific expansion was found in group XII among *R. communis* *V. fordii* and

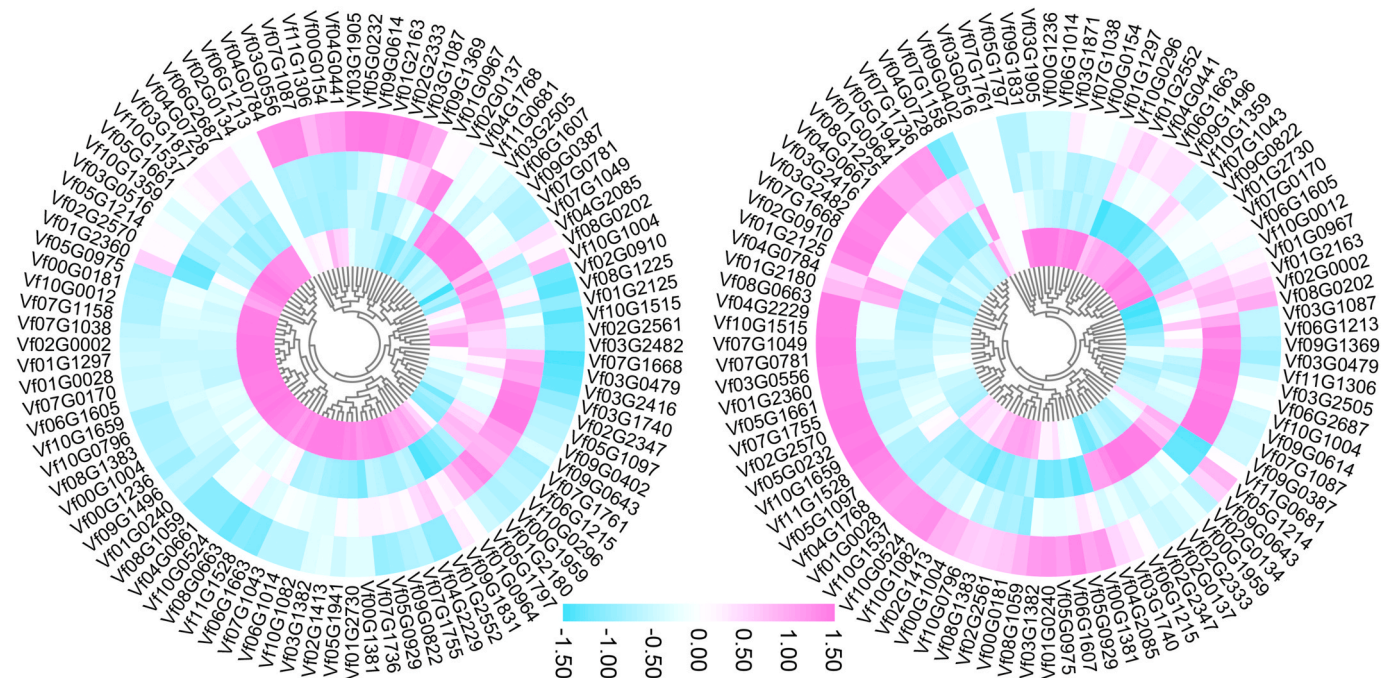


Fig. 5. Expression of *VjLRR-RLK* gene family members. The heat map depicts expression profiles of *VjLRR-RLKs* in *V. fordii* (left) and *V. montana* (right) in response to *Fusarium* wilt at four infection stages: 0, uninfected stage; 1, 2 days after *Fusarium* wilt infection (dpi); 2, 8 dpi; 3, 13 dpi. F0–F3 indicated the expression of *VjLRR-RLKs* in *V. fordii* during the infection stage (0, 1, 2, 3) by the pathogen *Fusarium* wilt; M0–M3 indicated the expression of *VjLRR-RLKs* in *V. montana* during the infection stage (0, 1, 2, 3) by the pathogen *Fusarium* wilt. The innermost circle represents 0, followed by 1, 2, and the outermost circle represents 3.

M. esculenta, which might be caused by the number of segmental duplications/WGDs making the number larger than other Euphorbiaceae species. Expression patterns according to the RNA-seq data suggested that most of the *VjLRR-RLKs* were expressed in a tissue-specific manner. Results from expression profiles demonstrated the resistance to *Fusarium* wilt function of *Vf01G2125*, *Vf03G1740*, *Vf06G2687*, and *Vf10G1659*. Our results open a new door to clarify the functions and evolutionary relationships of *LRR-RLK* genes in Euphorbiaceae, explore their roles in responding to *Fusarium* wilt infection and contribute to reveal the genetic mechanisms of resistance to *Fusarium* wilt infection and marker-assisted breeding in *V. fordii*.

4. Experimental

4.1. Database searches

Predicted genomes, proteomes, and genome annotation files from *J. curcas* (<http://www.kazusa.or.jp/jatropha/>), *M. esculenta* (<http://plants.ensembl.org/>), *H. brasiliensis* (<https://www.ncbi.nlm.nih.gov/>), *R. communis* (<https://phytozome.jgi.doe.gov/pz/portal.html>), *A. thaliana* (<https://www.arabidopsis.org/>), and *V. fordii* (<http://www.ncbi.nlm.nih.gov/genbank>) were collected and then downloaded. Subsequently, we used the InterProScan to determine different protein signatures in these predicted datasets (Jones et al., 2014). The local relational databases were developed to analyze and recover the dataset in each included plant genome. By searching for *LRR-RLKs* with leucine-rich repeat, kinases, and transmembrane domains, we retrieved *LRR-RLK* homologs in each relational database. Based on the Pfam (Punta et al., 2011), the Pkinase (PF00069) and Pkinase_Tyr (PF07714) were considered as diagnostic domains to recover the identifiers with KD. The TMHMM website was used to predict TMs. The *LRR* diagnostic domains, including *LRRNT_2* (PF08263), *LRRCT* (PF01463), *LRRV* (PF01816), *LRRNT* (PF01462), *LRR_1* (PF00560), *LRR_2* (PF07723), *LRR_3* (PF07725), *LRR_4* (2 copies) (PF12799), *LRR_5* (6 copies) (PF13306), *LRR_8* (PF13855), and *LRR_9* (PF14580), were searched. Finally, Pfam (E-value < 0.001) (Punta et al., 2011), InterProScan

(E-value < 0.001) (Jones et al., 2014) and CDD (E-value < 0.001) (Marchler-Bauer et al., 2012) were used to confirm the putative proteins. In our study, if the putative sequence contains KD, LRRs, and TM, they are considered as *LRR-RLK*.

4.2. Phylogenetic analysis

All full-length *LRR-RLK* proteins were subjected to MAFFT (Katoh and Standley, 2013) performing alignment by default settings. The different evolutionary models were tested using ModelFinder software (Kalyaanamoorthy et al., 2017). Maximum-likelihood (ML) tree was carried out with an SH-aLRT test (1000 random addition replicates) and a bootstrap test (10,000 replicates), as implemented in IQ-tree software (Nguyen et al., 2014). The phylogenetic trees were edited and visualized using the iTOL website and FigTree software (Letunic and Bork, 2016).

4.3. Synteny analysis and chromosomal distribution

The MCScanX (Wang et al., 2012) was used to identify the synteny relationships among genome regions with an E-value of $1e^{-10}$. The “add_ka_and_ks_to_collinearity.pl” was used to survey and estimate the synonymous and non-synonymous rates, and the different types of duplications (tandem, dispersed, proximal, and WGD or segmental) in the MCScanX package. The synteny relationships were visualized by Circos and TBtools software (Chen et al., 2020), as described by (Cao et al., 2020b).

4.4. Transcriptome expression analysis

The transcriptome data were collected and retrieved from the NCBI database (PRJNA445068, PRJNA483508, and PRJNA318350) to analyze the expression patterns. The SRA toolkit was used to decompress raw data into the FASTQ format. Then we use the HISAT2 to map each dataset into its corresponding template genome with default parameters (Kim et al., 2015; Pertea et al., 2016). The StringTie was used to determine the expression level from fragments per kilobase of exon per

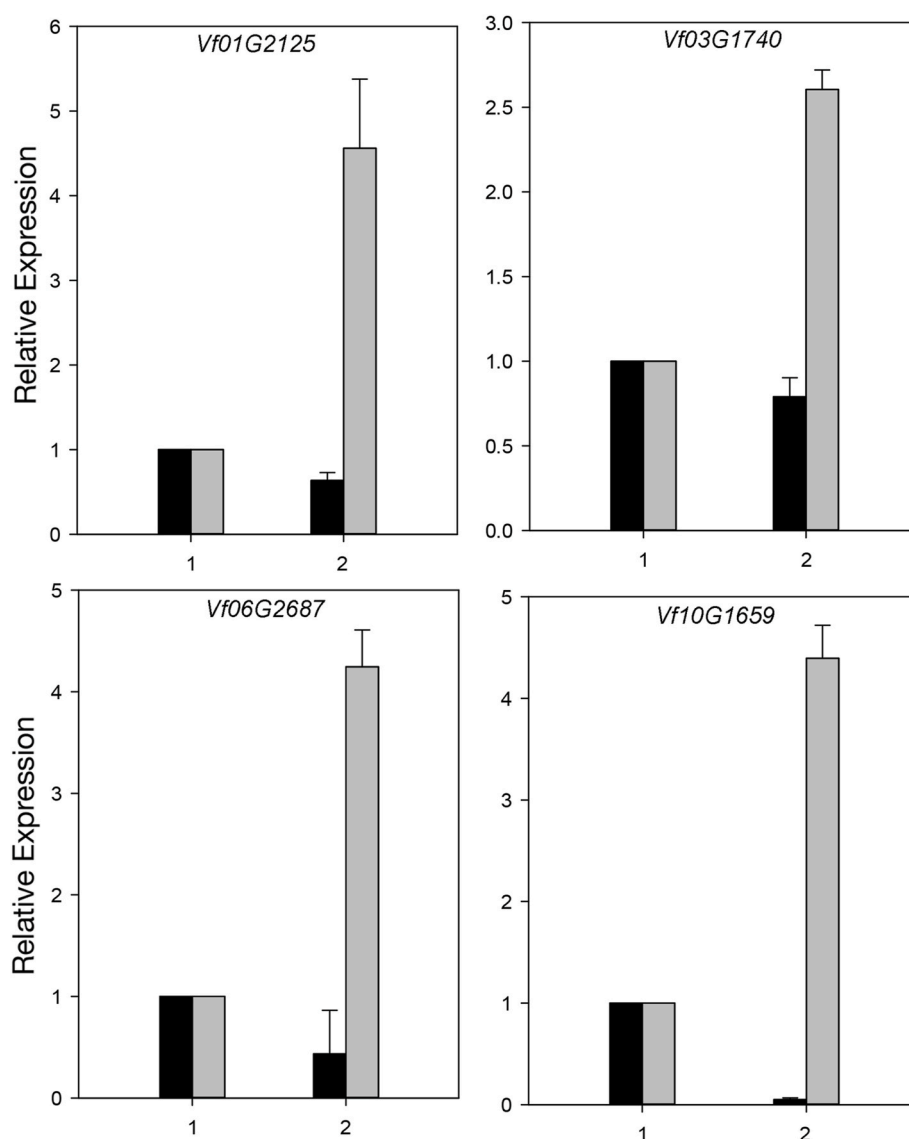


Fig. 6. The qRT-PCR experiments of four *VfLRR-RLKs* in response to *Fusarium* wilt. Black represents these *VfLRR-RLKs* in response to *Fusarium* wilt in *V. fordii*. Grey represents these *VfLRR-RLKs* in response to *Fusarium* wilt in *V. montana*. The numbers in the x-axis indicate the two stages of infection, as follows: 1, uninfected stage; 2, late stage of infection.

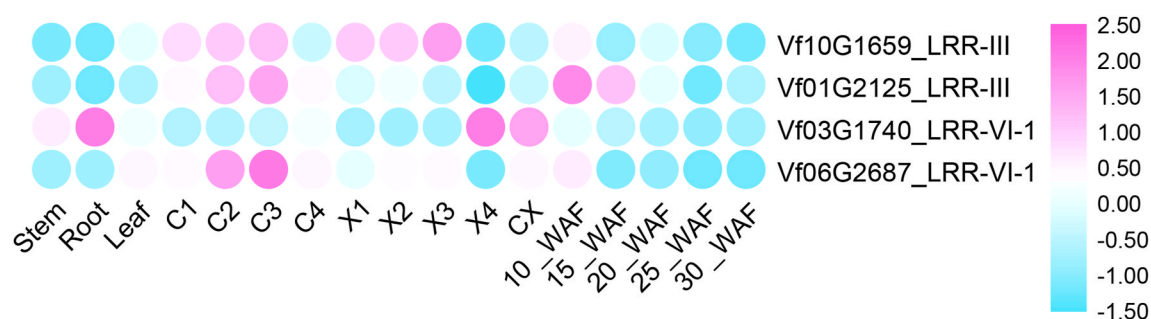


Fig. 7. The expression patterns of two gene pairs (i.e. *Vf01G2125* and *Vf03G1740*, and *Vf06G2687* and *Vf10G1659*) were generated by tandem duplication events. Numbers on the x-axis indicate the following: seeds at 10, 15, 20, 25, and 30 weeks after flowering (WAF; 10_WAF, 15_WAF, 20_WAF, 25_WAF, and 30_WAF), male flowers at 30, 20, 10, and 1 days before flowering (X1-X4), female flowers at 30, 20, 10, and 1 days before flowering (C1-C4), young leaves, roots, stems, and hermaphrodite (CX).

million reads (FPKM) with the default parameters (Pertea et al., 2016).

4.5. qRT-PCR experiment

Based on the non-conserved regions of *LRR-RLKs*, we designed the specific primers using Primer-BLAST on NCBI. The qRT-PCR experiment was carried out with the TransStart® Tip Green qPCR SuperMix (TransGen Biotech, China) and CFX96 Touch™ Real-Time PCR Detection System (Bio-Rad, USA) in a 20-μl reaction volume. The *VfEF1-α* gene (forward primer: 5'-GCCTGGTATGGTTGTGACCT-3'; reverse primer: 5'-GGATCATCCTTGGAGTTGGAC-3') was used as a reference gene (Liu et al., 2019), and the $2^{-\Delta\Delta C_t}$ method was used to estimate the relative expression levels of selected *LRR-RLKs*. Three biological repeats were executed for each sample.

4.6. Plant materials

V. montana Lour. and *V. fordii* (Hemsl.) Airy Shaw (Euphorbiaceae) are two species of *Vernicia* that were used as the plant materials in our study. The *Fof-1* was obtained from Anhui Agricultural University. The method of pathogen *Fof-1* inoculation is described by previously published papers (Chen et al., 2016; Zhang et al., 2018). Finally, we selected two stages which included untreated *Vernicia* species (*V. montana* and *V. fordii*) and *Vernicia* species (*V. montana* and *V. fordii*) infected with *Fusarium* wilt in the late stage (10 days after *Fusarium* wilt infection). For the study, fresh tissues of *V. fordii* were collected. Male flowers were sampled at 30 DBF (days before flowering: X1), 20 DBF (X2), 10 DBF (X3), and 1 DBF (X4). Female flowers were collected at 30 DBF (days before flowering: C1), 20 DBF (C2), 10 DBF (C3), and 1 DBF (C4). Young leaves, roots, stems, hermaphrodite (CX), and 30 WAF (weeks after flowering: 30_WAF) were collected in the spring.

Declaration of competing interest

The authors declare that there are no conflicts of interest.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.phytochem.2021.112686>.

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